

Velox Foods: AI-Powered Customer Review Analysis

Executive Summary

Recommendation: Allocate an estimated budget of **\$35.11** to process **50,000 customer reviews** using the **Groq Cloud API**.

The evaluation shows that while Ollama provides stronger data privacy, it delivers lower accuracy and slower processing. In contrast, Groq offers significantly higher accuracy, better coverage of customer feedback, and much faster performance.

Although the API introduces a small variable cost, it reduces processing time from **3.8 days to 5.6 hours** and produces more reliable insights. This enables faster decision-making and reduces the operational cost of analyst time tied up in processing.

Conclusion:

Groq is the preferred solution when prioritizing speed, accuracy, and business impact over minimal processing cost and local-only data handling.

1. Objective

Compare a local LLM (Ollama) and a cloud LLM (Groq) for aspect-based sentiment analysis of customer reviews.

The goal is to identify the key drivers of declining customer satisfaction and stagnant franchise growth by automatically extracting customer opinions on :

- Food Quality
- Service
- Price
- Ambiance
- Overall rating

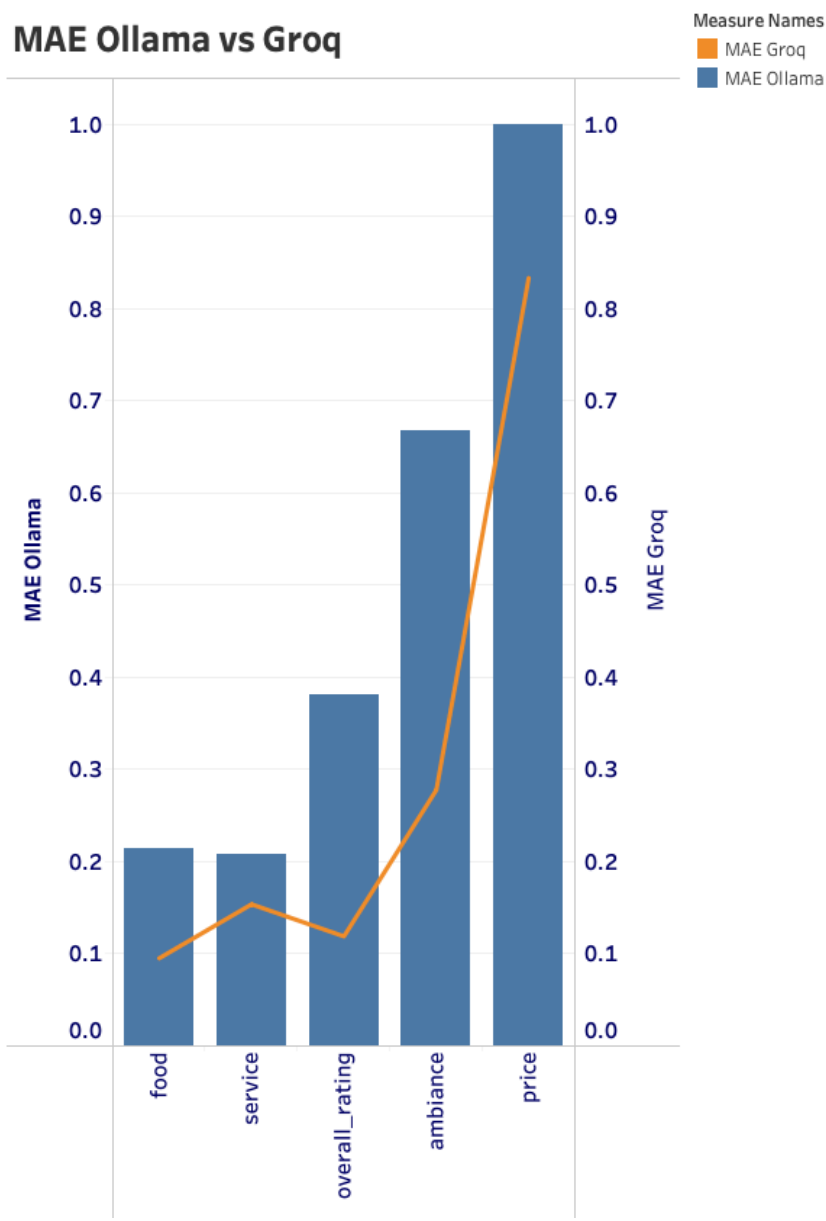
2. Models Evaluated

Two AI solutions were tested using the same dataset of 200 customer reviews:

	Ollama (Local)	Groq API (Cloud)
Privacy	Strong privacy control Data stays local	Data is processed externally via cloud infrastructure
Cost	Free software	Usage-based cost (API pricing)
Processing speed	Slow	Very fast
Prediction	Lower accuracy	Higher accuracy and more reliable results

3. Quantitative Results

Reliability (Mean Absolute Error)



Lower MAE indicates higher prediction reliability.

Business Interpretation

Groq consistently produced smaller prediction errors across every business category.

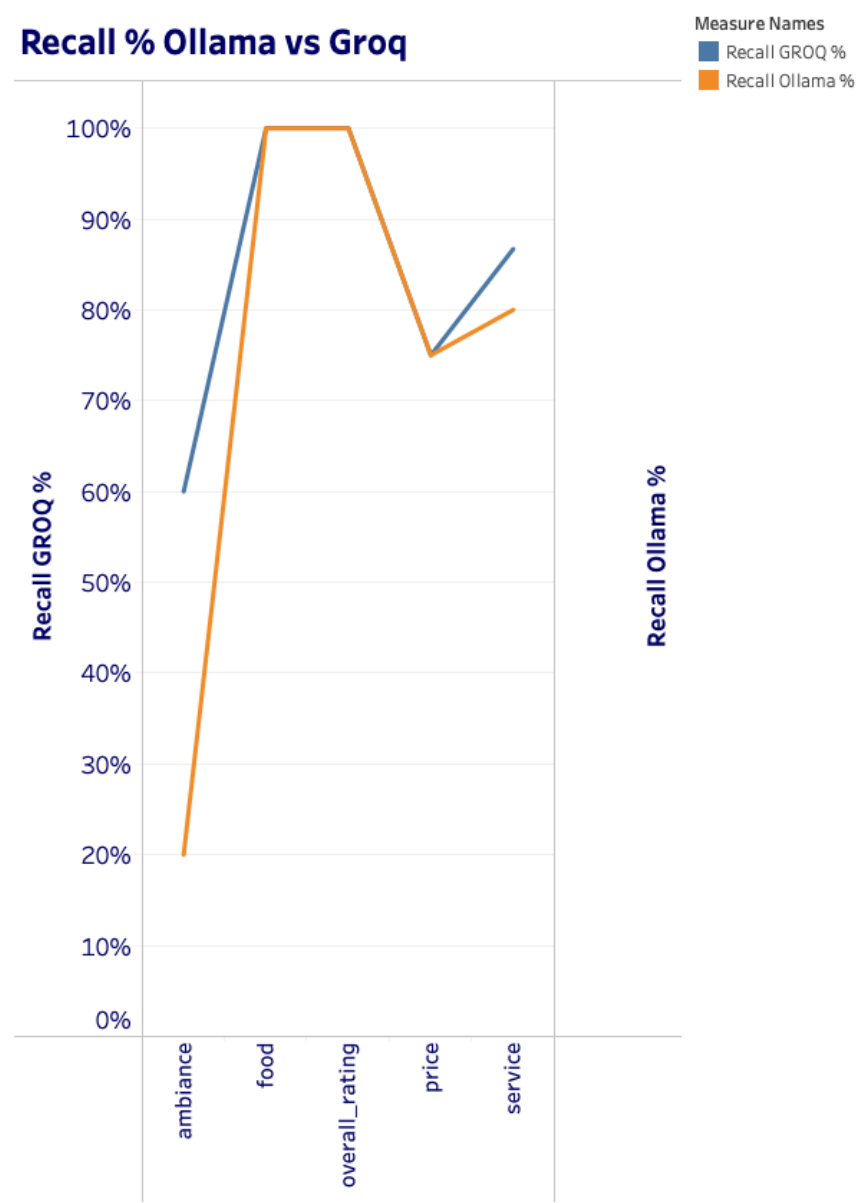
For example:

Customer ratings for **Food** were typically only **0.10 points away** from human evaluation, while Ollama's error was more than twice as large.

AVG MAE:
Ollama 0.494
Groq 0.296 (**ca. 40% lower error**)

More reliable predictions ensure that management decisions are based on accurate customer insights, leading to better prioritization of improvement initiatives.

Coverage (Recall)



Business Interpretation

Coverage represents operational visibility.

The most significant difference appears in **Ambiance**:

Ollama detected only **20%** of customer comments. Groq detected **60%**.

This means the local model ignores **4 out of every 5 customer comments** related to restaurant atmosphere.

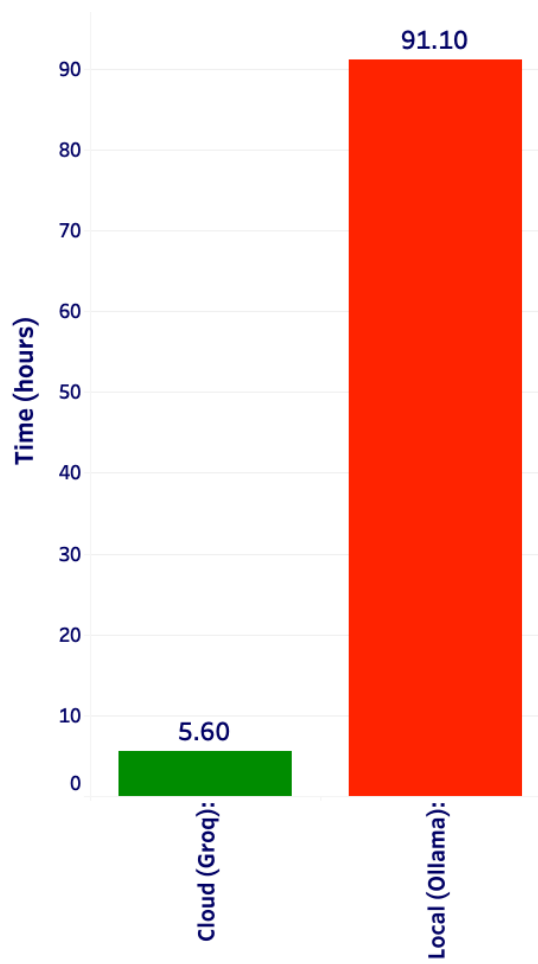
Average Recall

Ollama: 75%
Groq: 84%

Groq also shows higher average recall, meaning it consistently captures more relevant customer feedback across all categories.

Recall reflects how much of the relevant customer feedback is successfully captured and analyzed.

Processing Speed



Business Interpretation

Groq reduces processing time from 3.8 days to 5.6 hours, enabling near real-time analysis of customer feedback.

This results in a total of 85.5 hours of time savings per 50,000 reviews, allowing faster operational decisions and eliminating multi-day delays in insight generation.

Instead of occupying internal resources for nearly four days, results are available within a single working day, significantly improving the speed at which management can respond to customer issues.

Cost vs Time Analysis

Cost vs Time



Business Interpretation

Although Ollama has no direct API cost, the processing time creates hidden operational costs:

- workstation occupied for nearly 4 days
- reduced analyst productivity
- slower business decision-making

\$35 investment converts a 4-day analytics cycle into same-day business intelligence, significantly improving decision speed and insight accessibility.

4. Qualitative Insights

The evaluation shows that the Ollama-based model often misclassifies clearly positive restaurant reviews as negative. This happens mainly because it relies too heavily on individual words rather than understanding full context.

A recurring issue appears in reviews where positive sentiment is expressed using domain-specific or ambiguous vocabulary.

For example, in the review:

"Seems like a must while in NO. Mostly I enjoyed experiencing a snapshot of the NO community as I waited for takeout to enjoy on hotel rooftop patio. Late breakfast — debris (shredded roast beef in juice) on top of grits with cheese. Apparently grape jelly is a conventional condiment."

The human label is positive (1.0), yet the model assigns a strongly negative score (-1.0). The word *"debris"* is correctly interpreted as negative in general English, but in this context it refers to a local New Orleans dish. This indicates a lack of contextual and cultural understanding.

Similarly, in strongly positive reviews such as:

"Very yummy breakfast. Pancakes were awesome and grits were fantastical! LOL"

The model still outputs a negative score. This suggests it does not reliably capture strong positive language or informal emotional signals such as *"awesome"* or *"LOL"*.

Overall, the model struggles with:

- **Context-dependent meaning** (e.g., *"debris"* as food vs. literal debris)
- **Informal emotional cues** (e.g., *"LOL"*, *"awesome"*, *"fantastical"*)
- **Multi-sentence sentiment aggregation**, where overall meaning is positive despite varied phrasing

In contrast, higher-performing systems such as Groq handle these cases more effectively because they better understand context, linguistic nuances, and overall meaning rather than focusing on individual words. This leads to more accurate and consistent sentiment classification, resulting in more reliable business insights.

5. Conclusion

The evaluation demonstrates that the **Groq Cloud API is the superior solution** for customer review analysis at Velox Foods.

Compared with the local Ollama model, Groq provides:

- significantly lower prediction error
- higher recall across business categories
- much better detection of customer feedback on ambiance and service
- approximately sixteen times faster processing
- low implementation cost of approximately **\$35** for 50,000 reviews

The combination of higher reliability and dramatically reduced processing time makes the cloud solution the most cost-effective option despite the API fee.

6. Next Steps

If Approved Today

- Deploy Groq API pipeline immediately; process full historical review dataset
- Re-score and classify all reviews (Food, Service, Price, Ambiance)
- Build validated datasets across all locations (Week 1–2)
- Launch Tableau dashboards (Week 2) for trends, issues, and location comparison
- Deliver first insights (Week 3) identifying underperforming locations and key drivers
- Establish monthly reporting cycle for Operations (Month 1 onward)

Outcome:

Fully operational customer intelligence system within 30 days enabling data-driven decisions at scale.

If Not Approved

- Analyze only selected high-impact locations or small review samples
- Use basic sentiment tagging only (positive / neutral / negative)
- Provide lightweight manual summary reports instead of dashboards
- Reassess full rollout when data volume or business need increases

Outcome:

Limited directional insights without scalable analytics or deep operational visibility.
